

Anamika Lochab

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EDUCATION

Purdue University, West Lafayette <i>PhD Computer Science</i> • Research Interests: Alignment and stable post-training of large language models using Reinforcement Learning, with an emphasis on probabilistic methods and uncertainty quantification for trustworthy AI. • Advisor: Ruqi Zhang	09/2024 – 05/2029 IN, USA
Rutgers University, New Brunswick <i>MS Computer Science</i> • Thesis: Fine-Tuning Large Language Models, Advisor: Yongfeng Zhang	09/2022 – 05/2024 NJ, USA
Vellore Institute of Technology, Bhopal <i>Bachelor of Technology in Computer Science and Engineering</i>	07/2018 – 07/2022 MP, India

PUBLICATIONS

- * denotes equal contribution
- [1] **VERA: Variational Inference Framework for Jailbreaking Large Language Models** *NeurIPS 2025*
Anamika Lochab*, Lu Yan*, Patrick Pynadath*, Xiangyu Zhang, Ruqi Zhang.
 - [2] **Energy-Based Reward Models for Robust Language Model Alignment** *COLM 2025*
Anamika Lochab, Ruqi Zhang.
 - [3] **Cascade Reward Sampling for Efficient Decoding-Time Alignment** *COLM 2025*
Bolian Li*, Yifan Wang*, Anamika Lochab*, Ananth Grama, Ruqi Zhang.
 - [4] **VERA-V: Variational Inference Framework for Jailbreaking Vision-Language Models** *ICML 2026*
Qilin Liao*, Anamika Lochab*, Ruqi Zhang.
 - [5] **Uniform-Correct Policy Optimization: Breaking RLVR's Indifference to Diversity** *Under Review*
Anamika Lochab, Bolian Li, Ruqi Zhang.
 - [6] **Addressing Performance Saturation for LLM RL via Precise Entropy Curve Control** *Under Review*
Bolian Li, Yifan Wang, Yi Ding Anamika Lochab, Ruqi Zhang.

RESEARCH EXPERIENCE

Uncertainty Quantification in Automatic Vial Inspection <i>Dr. Ruqi Zhang, Purdue University</i>	09/2025 – Present
Certified Robustness to AIP attacks in Deep Learning based Recommenders <i>Dr. Hao Wang, Machine Learning Group, Rutgers University</i> • Developed an approach to certify robustness against Adversarial Item Promotion (AIP) attacks on deep learning based recommender systems using auxiliary information i.e, textual item descriptions. Adapted continuous optimization techniques to introduce perturbation in embedding layers for gradient descent over discrete data.	04/2023 – 12/2023
Bias Mitigation in Large Language Models <i>Dr. Yongfeng Zhang, The Wise Lab, Rutgers University</i> • Engineered a multi-faceted adaptation strategy encompassing fine-tuning, prompt engineering, and instruction tuning to substantially elevate fairness and mitigate inherent biases in Large Language Models (LLMs).	08/2023 – 04/2024

WORK EXPERIENCE

Microsoft, Office of Applied Research <i>Research Intern</i>	05/2026 – 08/2026
SmartBridge Educational Services Private Limited <i>Data Analyst Internship</i> • Created a website to detect and predict e-payment phishing websites.	07/2021 – 08/2021
Excavate Research and Analysis <i>Summer Internship</i> • Worked on dynamic real-time web development in Angular and implemented statistical weighting and data analysis for various clients like Orange mobile company, VEON Bangladesh, and Banglalink.	04/2019 – 06/2019

TEACHING EXPERIENCE

Purdue University

2024 - 2025

CS 251: Data Structures and Algorithms

Rutgers University

2023 - 2024

CS 206: Intro to Discrete Structures II · CS 210: Data Management for Data Science · CS 439: Intro to Data Science

AWARDS AND HONORS

- NSF ACCESS Discover Project Award, 2025-2026
- NeurIPS Scholar Award and Travel Award, 2025
- COLM Travel Award, 2025
- Outstanding Project Award, Rutgers University, 2024
- Gold Medalist, Vellore Institute of Technology, Bhopal, 2022

ACADEMIC SERVICE (REVIEWER)

- NeurIPS 2026
- COLM 2026
- ICML 2026
- ACL Demo 2026
- ARR February 2025